A Multi-Sink Sandpile Torsor and Fiber-Collapsed Markov Chains for Sampling Balanced Graph Partitions

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Abstract

Sampling connected, population-balanced k-partitions of a graph, the mathematical core of political redistricting analysis, currently rests on Markov chains with no known mixing-time bounds; rigorous guarantees exist only for *unconstrained* spanning-forest walks. We make two contributions toward closing this gap. First, we develop the algebraic infrastructure of balanced-forest chains. We prove that the lattice quotient $\mathrm{Pic}_k(G) = \mathbb{Z}^{n-k}/\operatorname{im}\Delta^{(S)}$ attached to a root set S of size k is the sandpile group of the graph G_S obtained by identifying S to a single sink, that spanning k-forests of G rooted at S are exactly the spanning trees of G_S, and hence, by rotor-routing on G_S, that $\mathrm{Pic}_k(G)$ acts freely and transitively on rooted k-forests of any graph. This yields exact (Markov-chain-free) uniform sampling of rooted k-forests and, via the quotient multigraph Q_ξ, an *exact within-fiber resampler* for the set of forests or trees compatible with a fixed partition ξ. Second, we prove a *fiber collapse lemma*. A chain that alternates exact fiber resampling with an arbitrary reversible partition-changing kernel mixes *exactly* as fast as its projection onto partition space; the within-fiber coordinate contributes nothing to the mixing time. This eliminates, provably and by construction, one of the two sources of slow mixing in all existing redistricting chains, and reduces the open problem of balanced-chain mixing to a spectral gap estimate for a projected chain on partitions alone. We formulate the remaining estimates as precise conjectures for grid graphs, and on $2 \times n$ strips we complete it. We classify the balanced connected partitions of the strip exactly for every k, by an interface coding with $\Theta_k(n^{k-1})$ states whose stationary law concentrates exponentially on narrow interfaces. We prove that the projected window-resampling chain is irreducible for every k, that every merged window splits with probability at least $1/10$, and, by a two-phase coupling driven by the concentration, that the full chain mixes. For every fixed k, the fiber-Gibbs chain on balanced k-forests of the $2 \times n$ strip mixes in $O(k \log k)$ steps, so an approximately uniform sample costs near-linear time. To our knowledge this is the first polynomial mixing-time bound for a Markov chain supported on balanced connected partitions. Exact diagonalization shows that the projected spectral gap is in fact $1/(k-1)$ asymptotically, independent of n.

[NOTE] *This draft merges the "Tier 2" package (torsor infrastructure, exact fiber resampling, irreducibility) with the "Tier 1" target (first mixing bound for a balance-restricted chain).*

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Everything labeled DONE has a complete proof in this file or a precise citation. Everything labeled MISSING is work to do; each MISSING box says exactly what is needed. The ledger in Appendix A lists all of them in one table. Remove all colored boxes, this note, and Appendix A before submission.

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1 Introduction

Let $G = (V, E)$ be a connected graph with $n = |V|$ vertices and $m = |E|$ edges, and let $k \geq 2$. A *connected k-partition* of G is a partition $\xi = \{\xi_1, \ldots, \xi_k\}$ of V into k nonempty classes (*districts*), each inducing a connected subgraph. A partition is *ε-balanced* if every district's size (more generally, vertex-weighted population) lies within a factor $(1 \pm \varepsilon)$ of n/k. The central algorithmic problem of quantitative redistricting analysis is to sample from a prescribed distribution on balanced connected k-partitions, most prominently the *spanning tree distribution*

$$\pi_{\mathrm{sp}}(\xi) \;\propto\; \prod_{i=1}^{k} t(G[\xi_i]),$$

where $t(H)$ denotes the number of spanning trees of H, which arises as the projection of the uniform distribution on k-component spanning forests. Chains in practical use (recombination [9], its reversible forest variants [3, 7], the cycle walk [10], the marked edge walk [16], and the balanced up-down walk [1]) are supported by increasingly strong irreducibility theorems and extensive experiments, but *no polynomial mixing-time bound is known for any Markov chain whose state space is the balanced set*, on any nontrivial graph family. The only rigorous mixing results in the area concern *unconstrained* forest walks: the up-down walk on the rank-$(n-k)$ truncation of the graphic matroid mixes in $O(m \log m)$ steps on every graph [2], and Cannon, Pegden, and Tucker-Foltz [5] showed that on $m \times n$ grids the balanced states occupy a $1/\mathrm{poly}$ fraction of the unconstrained space, so that rejection from the unconstrained sampler runs in expected polynomial time. Whether a chain *living inside* the balanced set can be proved to mix rapidly is open; it is stated as the outstanding problem in [10, 1, 5]. This paper resolves it for $2 \times n$ strips, for every fixed number of districts.

Every chain on balanced forests suffers from two distinct sources of slow mixing, and existing analyses cannot separate them:

(S1) **within-fiber shuffling:** states carrying the same partition ξ (differing in within-district trees, or in which boundary edges the spanning structure uses) must be mixed among themselves; empirically this consumes the majority of the steps of tree-based chains [16, 10];

(S2) **partition-space geometry:** the projected walk on balanced partitions themselves must overcome the combinatorial geometry of the balanced set, the genuinely hard part, where torpidity examples exist [7] and irreducibility can even fail [1].

Contributions. This paper removes (S1) *exactly*, by algebraic means, and isolates (S2) as a clean projected-chain problem.

(C1) **A multi-sink sandpile torsor (Theorem 3.1).** For any root set $S = \{q_1, \dots, q_k\} \subseteq V$, the lattice quotient $\mathrm{Pic}_k(G) := \mathbb{Z}^{n-k} / \operatorname{im} \Delta^{(S)}$ is canonically the sandpile (Picard) group of the multigraph G_S obtained by identifying S to a single sink; spanning k-forests of G rooted at S are exactly the spanning trees of G_S; and rotor-routing on G_S makes the set of rooted k-forests a *torsor* (free transitive action) for $\mathrm{Pic}_k(G)$, on every connected graph, with no planarity hypothesis. Consequences: exact uniform sampling of rooted k-forests by Wilson's algorithm on G_S (no Markov chain, no mixing bound), a multi-sink Dhar burning algorithm, and polynomial-time navigation of forest space.

(C2) **Exact fiber resampling (Theorem 4.1, Proposition 4.2).** The set of spanning trees compatible with a fixed partition ξ factorizes as $\prod_i \mathcal{T}(G[\xi_i]) \times \mathcal{T}(Q_\xi)$, where Q_ξ is the quotient multigraph; both factors admit exact uniform samplers (Wilson within districts; Wilson or the $\mathrm{Pic}^0(Q_\xi)$-torsor on the quotient). This gives a polynomial-time *exact* sampler of the conditional distribution given ξ, for both forest-valued and tree-valued state spaces.

(C3) **The fiber collapse lemma (Lemma 5.1, Theorem 5.2).** If a chain alternates the exact fiber resampler with an arbitrary π-reversible partition-changing kernel P_M, then the induced partition process is itself Markov, the joint chain's distance to stationarity

equals that of the projected chain, and the spectrum of the joint kernel is the spectrum of the projected kernel together with 0. With exact fiber resampling, the mixing problem for balanced forests *is* the mixing problem for balanced partitions, and (S1) is gone.

(C4) **The first mixing bound for a balanced-partition chain (Section 6).** On $2 \times n$ strips we classify the balanced partitions exactly for every k (Proposition 6.8), prove irreducibility of the projected merge–split chain for every k (Corollary 6.10), and prove that the fiber-Gibbs chain mixes in $O(k \log k)$ steps (Corollary 6.4). The collapse lemma reduces the problem to a heat-bath interface chain, which a concentration-driven coupling mixes in two sweeps. Irreducibility on rectangles (Conjecture 6.1) and mixing on two-dimensional grids remain open.

(C5) **Machine verification (Section 7).** All algebraic results were verified exhaustively on small graphs, including the torsor axioms on multigraph quotients, and we report an exhaustive irreducibility computation for a boundary-restricted chain on the 3×3 grid. We also report a computation showing that partition fibers are *not* cosets of subgroups of $\mathrm{Pic}_k(G)$, evidence that the collapse route of (C3), rather than a group-theoretic treatment of balance, is the right division of labor.

[DONE] *The introduction is essentially complete; the related-work paragraphs below may need expansion after the missing results stabilize.*

1.1 Overview of the argument

Sampling a balanced districting plan amounts to running a random walk on the balanced k-forests of a graph. Two separate effects slow every walk of this kind. Forests that induce the same district map differ in their within-district trees, and a walk spends most of its steps reshuffling those trees. Beyond that, the walk has to move between maps, and the geometry of the balanced maps may itself be difficult to cross.

We treat the two effects separately. The first is removed exactly. Within a fixed map the within-district trees correspond to the spanning trees of the graph obtained by collapsing each district to a point, and the sandpile group of the collapsed graph acts on them freely and transitively (Sections 3 and 4). The hidden part of the state can therefore be redrawn exactly, in one step, by Wilson's algorithm. The fiber collapse lemma of Section 5 shows that a chain which redraws the hidden part exactly mixes precisely as fast as its projection onto maps, so the entire problem reduces to the projected walk.

We then analyze the projected walk in a setting where it can be understood completely, the $2 \times n$ strip. A balanced map of the strip is described by the widths of the $k-1$ interfaces between consecutive districts, and the target distribution favors narrow interfaces, at a cost of roughly $\lambda = 2 + \sqrt{3}$ per additional cell of width (Section 6.3). Redrawing one interface is consequently a heavily biased draw, and two coupled copies of the walk agree after each interface has been redrawn twice (Section 6.4). This yields the first Markov chain on balanced partitions with a proven polynomial mixing time (Corollary 6.4). Square grids remain open, and Section 7 records the estimates a two-dimensional proof would require.

1.2 Related work

Torsor structures of the sandpile group on *spanning trees* originate with rotor-routing [13] and the Bernardi process [4]; on plane graphs the two coincide and are canonical [4, 6], and rotor-routing is the *only* consistent torsor algorithm there [12]. Off planarity both actions depend on the base data [18]. Our Theorem 3.1 is a reduction to [13]; its value is the systematic transfer of the tree-level theory to rooted k-forests and the resulting samplers. On the redistricting side, the irreducibility state of the art is [1] (exact balance, $k = 2$, simple grid graphs, and triomino districts on rectangles); the mixing state of the art for unconstrained forests is [2]; balanced sampling on grids by rejection is [5]; and the enumerative structure of balanced trees of $2 \times n$ strips is [11]. The decomposition method for Markov chains goes back to [14]; our Lemma 5.1 is the degenerate (and, for our purposes, optimal) case in which the restriction chains are exact resamplers, so that the decomposition becomes an identity rather than an inequality.

2 Preliminaries

A handful of symbols carry the paper: n and k (the strip has $2n$ cells and k districts), $s = 2n/k$ (cells per district), $t(H)$ (the number of spanning trees of a graph H), and $\lambda = 2 + \sqrt{3}$ (the growth rate of t on strips). Everything else is introduced where it is first used.

2.1 Graphs, forests, partitions

All graphs are finite and connected; multigraphs (parallel edges) are allowed and arise naturally as quotients; loops are deleted whenever they arise. For $U \subseteq V$, $G[U]$ is the induced subgraph. A *spanning k-forest* is an acyclic edge set $F \subseteq E$ with exactly k connected components whose vertex classes we denote $\xi(F) = \{\xi_1, \dots, \xi_k\}$; then $|F| = n - k$ and every ξ_i induces a connected subgraph, so $\xi(F)$ is a connected k-partition. Conversely the set of k-forests inducing a fixed ξ is $\prod_i \mathcal{T}(G[\xi_i])$, where $\mathcal{T}(H)$ denotes the set of spanning trees of H; hence the uniform distribution on k-forests projects to $\pi_{\mathrm{sp}}(\xi) \propto \prod_i t(G[\xi_i])$ on connected k-partitions.

Given a connected k-partition ξ, the *quotient multigraph* Q_ξ has one vertex per district and one edge per *boundary edge* of G (an edge with endpoints in two different districts), with multiplicities retained. A spanning tree $T \in \mathcal{T}(G)$ is *compatible* with ξ if $T \cap E(G[\xi_i])$ is a spanning tree of $G[\xi_i]$ for every i; in that case T contains exactly $k - 1$ boundary edges and these project to a spanning tree of Q_ξ.

Fix a nonnegative vertex weight (population) $p : V \to \mathbb{R}_{\geq 0}$, with $p \equiv 1$ unless stated otherwise, and $\bar{p} = p(V)/k$. A connected k-partition is *ε-balanced* if $|p(\xi_i) - \bar{p}| \leq \varepsilon \bar{p}$ for all i; "exact balance" is $\varepsilon = 0$. Write $\mathcal{B}_k^\varepsilon(G)$ for the set of ε-balanced connected k-partitions and $\mathcal{F}_k^\varepsilon(G)$ for the set of k-forests F with $\xi(F) \in \mathcal{B}_k^\varepsilon(G)$.

2.2 Divisors, the Picard group, and reduced Laplacians

$\mathrm{Div}(G) = \mathbb{Z}^V$ is the group of *divisors* (chip configurations); Δ_G is the graph Laplacian, acting on $\mathbb{Z}^V$ with $\Delta_G(v, v) = \deg v$ and $\Delta_G(u, v) = -(\#\text{ edges } uv)$. Divisors $D \sim D'$ are *linearly equivalent* if $D - D' \in \operatorname{im} \Delta_G$. The *Picard (sandpile, critical) group* of a connected multigraph H is $\mathrm{Pic}^0(H) = \mathrm{Div}^0(H)/\operatorname{im} \Delta_H$, where Div^0 is the degree-zero subgroup; by the matrix-tree

theorem $|\operatorname{Pic}^0(H)| = t(H)$. For $W \subseteq V$ the matrix $\Delta^{(W)}$ is Δ_G with the rows and columns of W deleted. The *Matrix-Forest theorem* (principal-minors form; see [8]) states that for $|W| = k$,

$$\det \Delta^{(W)} \;=\; \#\{\text{spanning } k\text{-forests of } G \text{ in which each component contains exactly one vertex of } W\}.$$

We call such forests *rooted at* W and write $\mathcal{F}_W(G)$ for their set. Following the April 2026 research notes we set

$$\operatorname{Pic}_k(G) \;=\; \mathbb{Z}^{n-k} \,/\, \operatorname{im} \Delta^{(S)}$$

for a fixed root set S of size k, so $|\operatorname{Pic}_k(G)| = |\mathcal{F}_S(G)|$.

2.3 Rotor-routing

Let H be a connected multigraph with a distinguished *sink* s, and fix a *ribbon structure*: a cyclic ordering of the edge-ends at each vertex. A *rotor configuration* assigns to each non-sink vertex v one of its incident edge-ends $\rho(v)$. Adding a chip at v and *rotor-routing* means: while the chip is at a non-sink vertex u, advance $\rho(u)$ to the next incident edge-end in the cyclic order and move the chip along that edge; stop when the chip reaches s. If the initial rotors $\{(v, \rho(v))\}$ form a spanning tree oriented toward s, so do the final rotors. The fundamental fact we use is:

Fact 2.1 (Holroyd–Levine–Mészáros–Peres–Propp–Wilson [13]; see also [12, Thm. 2.18])**.** *For every connected multigraph H with sink s and every ribbon structure, the rule "add a chip at v and rotor-route to the sink" induces a well-defined action of $\operatorname{Pic}^0(H)$ on $\mathcal{T}(H)$ which is free and transitive.*

2.4 Markov chain conventions

All chains are on finite state spaces. A kernel P is reversible with respect to π if $\pi(x)P(x,y) = \pi(y)P(y,x)$; then P is self-adjoint on $L^2(\pi)$ with real spectrum $1 = \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_{|\Omega|} \geq -1$. The *spectral gap* is $\gamma(P) = 1 - \lambda_2(P)$, and $t_{\mathrm{mix}}(P, \epsilon) = \min\{t : \max_x \left\|P^t(x,\cdot) - \pi\right\|_{\mathrm{TV}} \leq \epsilon\}$.

[DONE] *Section 2 is complete and self-contained.*

3 The multi-sink sandpile torsor

Throughout this section fix a root set $S = \{q_1, \ldots, q_k\} \subseteq V$, $2 \leq k < n$. Let G_S be the multigraph obtained from G by identifying all vertices of S to a single vertex s^* and deleting the loops created by edges with both endpoints in S. Edges of G not inside S correspond bijectively to edges of G_S; we use this identification of edge sets silently.

Theorem 3.1 (Multi-sink torsor)**.** *Let G be connected and S as above.*

(i) $\Delta^{(S)}$ *equals the reduced Laplacian of G_S at s^*. In particular* $\operatorname{Pic}_k(G) \cong \operatorname{Pic}^0(G_S)$*, an isomorphism of explicitly presented groups (identical integer matrices, hence identical Smith normal forms), and* $|\operatorname{Pic}_k(G)| = t(G_S)$.

(ii) The map $F \mapsto F$ (on edge sets) is a bijection $\mathcal{F}_S(G) \to \mathcal{T}(G_S)$ between spanning k-forests of G rooted at S and spanning trees of G_S.

(iii) For every ribbon structure on G_S, rotor-routing to the sink s^* defines a free and transitive action of $\mathrm{Pic}_k(G)$ on $\mathcal{F}_S(G)$.

Informally, rooted k-forests of G are the spanning trees of the graph obtained by identifying the k roots, and the sandpile group of that graph permutes them freely and transitively. A uniform rooted forest can therefore be drawn exactly, and no Markov chain is needed.

Proof. (i) The vertices of G_S other than s^* are exactly $V \setminus S$. For $u, v \in V \setminus S$, identification changes neither the multiplicity of edges between u and v nor $\deg u$: every edge from u into S becomes an edge from u to s^*, and no incident edge of u is deleted (only edges inside S become loops). Hence the reduced Laplacian of G_S at s^*, the matrix indexed by $V \setminus S$ with diagonal $\deg_G(u)$ and off-diagonal $-(\#$ edges uv in $G)$, coincides entry by entry with $\Delta^{(S)}$. The group isomorphism and the counting consequence follow from the definitions and the matrix-tree theorem for multigraphs.

(ii) Let $F \in \mathcal{F}_S(G)$. No edge of F joins two vertices of S: such an edge would place two roots in one component. Hence F is a well-defined edge set of G_S. It has $n-k$ edges and G_S has $n-k+1$ vertices, so it suffices to show the image is connected and acyclic. Connected: identifying the k roots glues the k component trees of F at the single vertex s^*. Acyclic: a cycle in the image either avoids s^*, lifting to a cycle of F in G (impossible), or passes through s^*, lifting to a path in F between two distinct roots (impossible: distinct roots lie in distinct components of F).

Conversely let $T \in \mathcal{T}(G_S)$ and let F be the corresponding edge set of G. F is acyclic: a cycle of F avoiding S maps to a cycle of T; a path of F between two distinct roots maps to a cycle of T through s^*; a cycle of F through one root maps to a cycle of T. Having $n-k$ edges on n vertices, F is a spanning k-forest. Every $v \in V \setminus S$ is connected in F to some root: the T-path from v to s^* lifts, edge by edge until its first edge incident to s^*, to an F-path from v to some q_i. So each of the k components of F contains at least one root, and by counting exactly one. The two maps are mutually inverse because both are the identity on edge sets.

(iii) Combine (i), (ii), and Fact 2.1 applied to $H = G_S$ with sink s^*. □

[DONE] *Theorem 3.1 is fully proved above. It was additionally verified by machine (freeness, transitivity, commutativity, and the chip-firing relations of the action) on the 2×3 grid with root sets of sizes 2 and 3, the 3×3 grid, and K_4 with an adjacent root pair, where G_S has parallel edges; see Section 7.*

Remark 3.2 (Multi-sink Dhar burning)**.** *Dhar's burning algorithm on G_S started at s^* is precisely the multi-source process on G in which fire spreads simultaneously from all k roots; the edges that carry the fire form the rooted k-forest, and the root whose fire claims a vertex defines a Voronoi-type assignment. All recurrence tests and reduced-divisor computations for the k-sink sandpile on G are ordinary single-sink computations on G_S.*

Remark 3.3 (Canonicity caveat)**.** *G_S is in general non-planar even when G is planar, so the action of Theorem 3.1(iii) depends on the ribbon structure and there is no canonical choice: off planarity both the rotor and Bernardi torsors are base-dependent [18], and on plane graphs*

rotor-routing is the only *consistent torsor algorithm [12]. None of the applications below require canonicity.*

Corollary 3.4 (Exact sampling and navigation)**.** *Fix S.*

(a) Running Wilson's algorithm [19] on G_S rooted at s^* returns a uniformly random element of $\mathcal{F}_S(G)$ in expected time proportional to the mean hitting time of G_S (in particular, expected polynomial time; $O(n \log n)$ on bounded-degree planar graphs such as grid quotients).

(b) Given the Smith normal form

$$U\Delta^{(S)}V = \mathrm{diag}(d_1, \dots, d_{n-k}),$$

a uniform element of $\mathrm{Pic}_k(G) \cong \prod_i \mathbb{Z}/d_i$ is sampled by choosing uniform coordinates, and its action on a stored reference forest is evaluated in polynomial time by decomposing a degree-zero representative into elementary chip additions and rotor-routing each chip to the sink.

Proof. (a) is Wilson's theorem applied to G_S, using Theorem 3.1(ii) to transport uniformity. (b) The SNF provides the isomorphism and a representative divisor for each coordinate vector; a degree-zero divisor D on G_S can be written $D = \sum_v D(v)\big((v) - (s^*)\big)$ with $\sum_v |D(v)|$ bounded polynomially after replacing D by its s^*-reduced representative (computable by Dhar's algorithm in polynomial time), and each elementary action $(v)-(s^*)$ is one rotor walk of at most $2|E|$ steps by [12, Prop. 3.3] when v is adjacent to s^*, and at most polynomially many steps in general [13]. □

[NOTE] *Corollary 3.4(b) is stated in the safe direction: we sample a group element and* apply *it. The inverse problem, that of computing from two forests the unique group element carrying one to the other under the rotor action (the torsor discrete logarithm), is not needed anywhere in this paper. If we later want it for canonical-path constructions, the clean route is to use any polynomial-time bijection $\mathcal{T}(G_S) \leftrightarrow \mathrm{Pic}^0(G_S)$ (e.g. Dhar's bijection through reduced divisors) instead of inverting the rotor action; verify carefully which statements need the action itself and which need only* some *bijection.*

4 Fibers and exact fiber resampling

Fix a connected k-partition ξ. Two state spaces occur in the literature: *forest-valued* chains store $F \in \mathcal{F}_k(G)$, and *tree-valued* chains store a compatible spanning tree (possibly with marked boundary edges). We treat both; the tree case is where the quotient group enters.

Theorem 4.1 (Fiber decomposition)**.** *For every connected k-partition ξ, the map*

$$T \longmapsto \Big((T \cap E(G[\xi_i]))_{i=1}^{k},\ T \cap \partial\xi \Big)$$

is a bijection between the compatible spanning trees of G and $\prod_{i=1}^{k} \mathcal{T}(G[\xi_i]) \times \mathcal{T}(Q_\xi)$. Consequently

$$\tau(\xi) := \#\{\textit{compatible spanning trees}\} = \Big(\prod_{i=1}^{k} t(G[\xi_i])\Big) \cdot t(Q_\xi),$$

$$\#\{k\textit{-forests with } \xi(F) = \xi\} = \prod_{i=1}^{k} t(G[\xi_i]).$$

Proof. If T is compatible, each $T \cap E(G[\xi_i])$ is by definition a spanning tree of $G[\xi_i]$, accounting for $\sum_i (|\xi_i| - 1) = n - k$ edges; the remaining $k - 1$ edges of T are boundary edges, and they connect the k districts without cycles (contracting each district maps T onto a connected, $(k-1)$-edge subgraph of Q_ξ), i.e. they form a spanning tree of the multigraph Q_ξ. Conversely, the union of spanning trees T_i of the districts and a spanning tree σ of Q_ξ (a choice of $k-1$ boundary edges of G connecting all districts acyclically) has $n-1$ edges, spans V, and is acyclic: a cycle would either live inside a district (impossible, T_i is a tree) or project to a nontrivial closed walk in the tree σ (impossible). The two constructions are mutually inverse. The forest statement is the case in which no boundary edges are selected. □

[DONE] *Theorem 4.1 is fully proved and was verified by brute-force enumeration on five partitions of the 2×3 and 3×3 grids and on a 6×6 instance (Section 7).*

Proposition 4.2 (Exact fiber resampler)**.** *Fix ξ. Each of the following runs in expected polynomial time and returns an exactly uniform sample:*

(a) a uniform k-forest with partition ξ: run Wilson's algorithm independently in each $G[\xi_i]$;

(b) a uniform compatible spanning tree: additionally sample a uniform spanning tree of the multigraph Q_ξ, either by Wilson's algorithm on Q_ξ, or by sampling a uniform $g \in \mathrm{Pic}^0(Q_\xi)$ via the Smith normal form and letting g act on a reference tree of Q_ξ through rotor-routing (Fact 2.1), and take the union as in Theorem 4.1.

Proof. Immediate from Theorem 4.1, Wilson's theorem, and, for the torsor route in (b), the fact that if the action of a finite group on a finite set is free and transitive then the orbit of any fixed point under a uniform group element is uniform. □

Remark 4.3. *The torsor route in (b) is stated because the group $\mathrm{Pic}^0(Q_\xi)$, not any particular sampler, is the structural object: it indexes the* boundary configurations *of ξ and quotients away the within-fiber redundancy that tree-valued chains otherwise spend most of their steps shuffling [16]. In implementations Wilson on Q_ξ is simpler and equally exact.*

5 The fiber collapse lemma

We now formalize the reduction (S1) ⇒ gone. The lemma is stated abstractly; the intended instance is $\Omega = \mathcal{F}_k^\varepsilon(G)$ (or the compatible-tree space), fiberd over $\Xi = \mathcal{B}_k^\varepsilon(G)$ by $x \mapsto \xi(x)$, with P_F the exact resampler of Proposition 4.2.

Setting. Let Ω be a finite set, $\zeta : \Omega \to \Xi$ a surjection onto a finite index set (*fibration*), and π a probability distribution on Ω with full support. Write $\bar{\pi} = \zeta_* \pi$ for the pushforward and $\nu_\xi = \pi(\cdot \mid \zeta = \xi)$ for the conditionals, so that $\pi = \sum_\xi \bar{\pi}(\xi)\, \nu_\xi$ and the ν_ξ have pairwise disjoint supports. Define the *fiber resampling kernel*

$$P_F(x, y) = \nu_{\zeta(x)}(y),$$

and let P_M be any transition kernel on Ω reversible with respect to π. Define the *sandwich kernel* $P = P_F P_M P_F$ (apply P_F, then P_M, then P_F) and the *projected kernel* on Ξ

$$\bar{P}(\xi, \xi') \;=\; \sum_{x \in \zeta^{-1}(\xi)} \nu_\xi(x) \sum_{y \in \zeta^{-1}(\xi')} P_M(x, y).$$

Lemma 5.1 (Fiber collapse)**.** *With the notation above:*

(a) P_F is reversible with respect to π, idempotent ($P_F^2 = P_F$), and as an operator on $L^2(\pi)$ equals the orthogonal projection $\Pi f = \mathbb{E}_\pi[f \mid \zeta]$ onto the subspace of ζ-measurable functions. Consequently $P = \Pi M \Pi$ (with M the operator of P_M) is reversible with respect to π.

(b) $\bar{P}$ is reversible with respect to $\bar{\pi}$, and for the P-chain $(x_t)_{t \geq 0}$ started at any x_0, the projected process $\xi_t = \zeta(x_t)$ is a Markov chain with kernel $\bar{P}$ started at $\zeta(x_0)$.

(c) For every $t \geq 1$ and every x_0,

$$\| \operatorname{Law}(x_t) - \pi \|_{\mathrm{TV}} \;=\; \| \operatorname{Law}(\xi_t) - \bar{\pi} \|_{\mathrm{TV}} \;=\; \left\| \bar{P}^{\,t}(\zeta(x_0), \cdot) - \bar{\pi} \right\|_{\mathrm{TV}}.$$

In particular $t_{\mathrm{mix}}(P, \epsilon) = t_{\mathrm{mix}}(\bar{P}, \epsilon)$ for every ϵ.

(d) $\operatorname{Spec}(P) = \operatorname{Spec}(\bar{P}) \cup \{0\}$ (as multisets, the eigenvalue 0 appearing with multiplicity $|\Omega| - |\Xi|$). In particular $\lambda_2(P) = \max\{\lambda_2(\bar{P}), 0\}$ and $\lambda_{\min}(P) = \min\{\lambda_{\min}(\bar{P}), 0\}$.

Informally, once the hidden coordinate is redrawn exactly at every step it no longer matters; the chain and its projection onto partitions are equally far from stationarity at every time.

Proof. (a) For x, y in the same fiber $\zeta^{-1}(\xi)$, $\pi(x) P_F(x, y) = \bar{\pi}(\xi) \nu_\xi(x) \nu_\xi(y)$ is symmetric in (x, y); for x, y in different fibers both sides vanish. Idempotence is the fact that resampling twice from the same conditional is the same as resampling once (ζ is unchanged by P_F). For the operator statement, $(P_F f)(x) = \sum_y \nu_{\zeta(x)}(y) f(y) = \mathbb{E}_\pi[f \mid \zeta = \zeta(x)]$, which is precisely the conditional expectation; conditional expectation is the orthogonal projection onto $L^2(\bar{\pi}) \subseteq L^2(\pi)$, the subspace of ζ-measurable functions. Then $P^* = P_F^* P_M^* P_F^* = P_F P_M P_F = P$ since P_F, P_M are self-adjoint.

(b) Reversibility of $\bar{P}$: for $g, h : \Xi \to \mathbb{R}$, $\langle \bar{P} g, h \rangle_{\bar{\pi}} = \langle M(g \circ \zeta), h \circ \zeta \rangle_\pi$ is symmetric because M is self-adjoint. Markov property: one step of P from x_t consists of $y \sim \nu_{\zeta(x_t)}$, then $z \sim P_M(y, \cdot)$, then $x_{t+1} \sim \nu_{\zeta(z)}$. The law of $\zeta(x_{t+1}) = \zeta(z)$ depends on the past only through $\zeta(x_t)$ (the draw y erases the position within the fiber), and equals $\bar{P}(\zeta(x_t), \cdot)$ by definition of $\bar{P}$.

(c) After the terminal P_F in step t, the conditional law of x_t given $(\xi_0, \dots, \xi_t)$ is exactly ν_{ξ_t}. Hence $\mathrm{Law}(x_t) = \sum_\xi \Pr[\xi_t = \xi]\, \nu_\xi$, while $\pi = \sum_\xi \bar{\pi}(\xi)\nu_\xi$. Because the ν_ξ have disjoint supports,

$$\|\mathrm{Law}(x_t) - \pi\|_{\mathrm{TV}} = \tfrac{1}{2} \sum_\xi \sum_{x \in \zeta^{-1}(\xi)} \left|\Pr[\xi_t = \xi] - \bar{\pi}(\xi)\right| \nu_\xi(x) = \tfrac{1}{2} \sum_\xi \left|\Pr[\xi_t = \xi] - \bar{\pi}(\xi)\right|,$$

which is the projected total variation distance; the second equality in the statement is (b). Taking maxima over starting states (every $\xi_0 \in \Xi$ is realized by some x_0) gives the equality of mixing times.

(d) $L^2(\pi) = \mathrm{ran}\,\Pi \oplus \ker \Pi$. On $\mathrm{ran}\,\Pi \cong L^2(\bar{\pi})$ we have, for $f = g \circ \zeta$: $Pf = \Pi M \Pi f = \Pi M f = (\bar{P}g) \circ \zeta$, so $P|_{\mathrm{ran}\,\Pi}$ is conjugate to $\bar{P}$. On $\ker \Pi$: $Pf = \Pi M \Pi f = \Pi M 0 = 0$. The spectral statement follows. □

[DONE] *Lemma 5.1 is fully proved above and is elementary; its value is that Proposition 4.2 makes P_F* exactly *implementable for balanced forests, which is what turns the classical decomposition inequality of [14] into an identity.*

Theorem 5.2 (Conditional mixing theorem)**.** *Let $\Omega = \mathcal{F}_k^\varepsilon(G)$ with $\zeta = \xi(\cdot)$, let π be any distribution of the product form $\pi(F) = \bar{\pi}(\xi(F))\, \nu_{\xi(F)}(F)$ with ν_ξ uniform on the fiber (e.g. π uniform on $\mathcal{F}_k^\varepsilon(G)$, for which $\bar{\pi} = \pi_{\mathrm{sp}}$ conditioned on balance), and let P_M be any π-reversible partition-changing kernel with each step implementable in polynomial time. Then the sandwich chain $P = P_F P_M P_F$ is implementable in expected polynomial time per step, reversible with respect to π, and satisfies*

$$t_{\mathrm{mix}}(P, \epsilon) = t_{\mathrm{mix}}(\bar{P}, \epsilon) \quad \textit{for all } \epsilon.$$

In particular, if the projected chain $\bar{P}$ on $\mathcal{B}_k^\varepsilon(G)$ is irreducible with spectral gap $\gamma(\bar{P}) \geq 1/\mathrm{poly}(n)$, then P mixes in polynomial time on the balanced forest space. The same statements hold verbatim for the compatible-tree state space using Proposition 4.2(b).

Proof. Combine Proposition 4.2 (implementability of P_F), Lemma 5.1(a)–(d), and the standard bound $t_{\mathrm{mix}}(\bar{P}, \epsilon) \leq \gamma(\bar{P})^{-1} \log\big((\min_\xi \bar{\pi}(\xi))^{-1} \epsilon^{-1}\big)$ for reversible chains (applied after the usual lazy modification if $\lambda_{\min}(\bar{P}) < 0$; laziness changes the gap by a factor 2 and survives the sandwich construction unchanged). □

[DONE] *Theorem 5.2 is proved in full,* conditionally on a gap bound for the projected chain. *It is the precise interface between the algebraic half of the paper (done) and the geometric half (open, Section 6).*

Remark 5.3 (Why not treat balance algebraically?)**.** *One might hope the balanced set is itself an orbit or coset for a subgroup of $\mathrm{Pic}_k(G)$, so that Section 3 alone would finish the problem. It is not: machine verification on the 2×3 grid shows that already the fiber of a single partition is in general* not *a coset of any subgroup of $\mathrm{Pic}_k(G)$, nor is the union of the balanced fibers (Section 7). The group sees the forest structure exactly and the balance structure not at all; Lemma 5.1 is the correct division of labor.*

6 The projected chain and the strip theorem

By Theorem 5.2 everything now depends on the projected kernel $\bar{P}$ on $\mathcal{B}_k^{\varepsilon}(G)$. This section fixes a concrete P_M, states the two conjectures whose resolution would complete the program, and records the proof plans and the known results supporting them.

6.1 The partition-changing kernel

We take P_M to be the *window resampling* kernel. From a forest F with partition ξ, choose an edge $\{i, j\}$ of Q_ξ uniformly at random and replace the forest on $H = G[\xi_i \cup \xi_j]$ by an exact draw from its conditional distribution given the rest of the state; if the window is disconnected, hold. Under the uniform distribution on balanced forests the conditional factors. First choose a balanced split $\{A, B\}$ of H with probability proportional to $t(G[A])\, t(G[B])$, then draw uniform spanning trees of A and of B by Wilson's algorithm. On strips the split distribution is explicitly computable, since Proposition 6.9 lists the $O(s)$ balanced splits and their weights are products of ladder counts, available from the three-term recurrence. P_M is a Gibbs block update of the window forest given the rest, hence reversible with respect to π.

[NOTE] *This kernel replaced an earlier tree-based definition (uniform spanning tree of the window, uniform balanced split edge, MH correction). The pre-submission audit found two problems with the tree-based route: the exact MH correction of the uniform-tree proposal is not computable in closed form, and every implementable variant conditions the tree through the current split, which under exact balance freezes the projected chain outright (Remark 6.11; machine-verified). The heat-bath kernel has neither problem on strips, and the collapse lemma is indifferent to the choice.*

6.2 Irreducibility (Conjecture A)

Conjecture 6.1 (Irreducibility, exact balance, all k)**.** *Let G be an $a \times b$ rectangular grid graph with $k \mid ab$. The projected balanced merge–split chain $\bar{P}$ is irreducible on $\mathcal{B}_k^0(G)$, with diameter $O(ab)$ moves.*

Status and plan. For $k = 2$ this follows for the BUD chain from [1, Thm. B.2] on all simple grid graphs (any prescribed part sizes, diameter $O(n)$); a transfer argument from BUD moves to merge–split moves is needed and appears routine, since every BUD move between exactly balanced states is realized by a merge–split of the two districts involved (via [1, Lemma A.3], retreeing is free inside our framework: the fiber coordinate is irrelevant after collapse). For $k \geq 3$ the natural strategy, proposed in [1] and currently open, is recursion: prove the $k = 2$ statement not just for simple grid graphs but for the district-pair subgraphs $G[\xi_i \cup \xi_j]$ that arise along the recursion; these are connected but can have holes and cut vertices, where counterexamples are known to exist in general [1, Lemma B.14]. The plan:

1. Prove Conjecture 6.1 for $k = 3$ on rectangles directly, by adapting the canonicalization argument of [1, App. B]: canonicalize to bands of columns, using merge–split of *adjacent* bands only (the union of two adjacent canonical bands is simply connected, avoiding the holes obstruction).

2. Identify the class of intermediate shapes closed under the recursion (candidate: "column-convex" district pairs) and prove the $k = 2$ statement for that class.

3. General k by induction, always merging a canonical extreme band with its neighbor.

[MISSING] *All three steps. Step 1 is a finite, technical extension of a published argument and is the right first target (∼weeks, not months). Note the competition risk: the authors of [1] state the $k > 2$ extension as their own ongoing work. Our differentiator is that we only need irreducibility for* merge–split *on rectangles, not for BUD on arbitrary grid subgraphs; check whether restricting the ambition this way shortens their argument materially.*

Corollary 6.10 proves the strip analogue of Conjecture 6.1 outright; the triomino irreducibility of [1, Thm. B.13] on rectangles and the 3×3 computation of Section 7 support the rectangular case.

[NOTE] *Empirical support: on the 3×3 grid with $k = 3$ and exact balance, the* partition-level *communication classes of a boundary-restricted chain cover all 10 balanced partitions (Section 7); and [1, Thm. B.13] proves the triomino case ($k = ab/3$) on all rectangles, which includes 3×3, $k = 3$.*

6.3 The main theorem for $2 \times n$ strips

Theorem 6.2 (Strip gap)**.** *Fix $k \geq 3$ with $k \mid 2n$, exact balance, and write $s = 2n/k$. The projected window-resampling chain $\bar{P}$ on $\mathcal{B}_k^0(2 \times n)$ satisfies, with absolute constants $c, C_0 > 0$,*

$$\gamma(\bar{P}) \;\geq\; \frac{c}{k \log(k+1)} \qquad \text{for } s \geq C_0 \log(k+1),$$

and $\gamma(\bar{P}) > 0$ for the finitely many smaller s (by Corollary 6.10). The proof is in Section 6.4.

Remark 6.3 (Why $k \geq 3$)**.** *For $k = 2$ the only window is the whole strip, so a single update redraws the entire partition from its exact conditional distribution; the chain mixes in one step and yields no interesting theorem. The substantive question, where updates are genuinely local, begins at $k = 3$.*

Corollary 6.4 (Strip mixing, unconditional)**.** *Fix $k \geq 2$ with $k \mid 2n$ and $s = 2n/k$. Take P_M to be the window resampling kernel of Section 6.1 and $P = P_F P_M P_F$ the fiber-Gibbs balanced chain on $\mathcal{F}_k^0(2 \times n)$. Then*

$$t_{\text{mix}}(P, 1/4) \;\leq\; 2(k-1)\big(\lceil \log(k+1) \rceil + 4\big) \qquad \text{for } s \geq C_0 \log(k+1),$$

and $t_{\text{mix}}(P, 1/4) \leq C_k$ for the finitely many remaining n. Each step runs in expected time $O(n \log n)$ measured in random walk moves, plus arithmetic on $O(n)$-digit integers, so for every fixed k the sampler produces an approximately uniform balanced k-forest in near-linear expected total time. To our knowledge this is the first mixing-time bound for a Markov chain whose states are balanced connected partitions (lifted to forests) of a nontrivial graph family. The proof is at the end of Section 6.4.

The balanced partition space of a strip. We settle the $k = 2$ structure completely; it is the template for the general interface coding. Index the vertices of the $2 \times n$ grid by $\{0,1\} \times \{0,\dots,n-1\}$ (row, column); *column* c is the pair $\{(0,c),(1,c)\}$. For a 2-partition $\{A,B\}$, call a column *pure-A* (resp. *pure-B*) if both its vertices lie in A (resp. B), and *mixed* otherwise; the *orientation* of a mixed column records which part holds its top vertex.

Proposition 6.5 (Classification of balanced 2-partitions of the strip)**.** *Let $n \geq 2$. The balanced connected 2-partitions of the $2 \times n$ grid are exactly the following, and there are exactly n of them:*

(a) the horizontal split {top row, bottom row};

(b) for n even, the vertical split after column $n/2 - 1$;

(c) for each $a \in \{1,\dots,\lceil n/2 \rceil - 1\}$ and each $\sigma \in \{\text{top}, \text{bottom}\}$, the staircase $S_{a,\sigma} = \{P_{a,\sigma}, V \setminus P_{a,\sigma}\}$*, where $P_{a,\sigma}$ consists of the a leftmost (full) columns together with the σ-row vertices of the middle $n - 2a$ columns.*

Proof. Each listed partition is balanced ($|P_{a,\sigma}| = 2a + (n - 2a) = n$) and has connected parts: the middle vertices of $P_{a,\sigma}$ form a horizontal path attached to its full block, and the complement is the left–right, top–bottom mirror image. The left–right mirror of $S_{a,\text{top}}$ coincides with $S_{a,\text{bottom}}$ *as an unordered partition*, so each unordered partition appears exactly once in the list, and the listed partitions are pairwise distinct. It remains to show there are no others.

Let $\{A,B\}$ be a balanced connected 2-partition, with a', b', m the numbers of pure-A, pure-B, and mixed columns. Every mixed column contributes one vertex to each part, so $n = |A| = 2a' + m$ and $n = |B| = 2b' + m$; hence $a' = b'$.

Step 1: no pure columns ⇒ horizontal split. If all columns are mixed and two adjacent columns had different orientations, A would meet both rows; a connected set meeting both rows must contain a vertical edge, i.e. a pure column, a contradiction. So all orientations agree and A is a full row.

Step 2: the pure-A columns are consecutive; likewise pure-B. Suppose pure-A columns $c < c'$ have a non-pure-A column strictly between them. Any path between vertices on opposite sides of column c passes through column c, which lies entirely in A; hence the connected set B has no vertices on both sides of c, nor of c'. Since some column strictly between c and c' contains a B-vertex, B lies entirely in the open "pocket" between c and c'. Because A is connected, some A-path crosses the pocket from column c to column c'; inside the pocket it cannot change rows (a row change uses a vertical edge with both endpoints in A, i.e. a pure-A column, and the pocket contains none), so one full row of the pocket lies in A. Every pocket column, being non-pure-A, then has exactly its other-row vertex in B, whence $|B| = (\text{pocket width}) \leq n - 2 < n$, contradicting balance.

Step 3: mixed columns lie between the two pure blocks, which sit at the two ends. If $a' = 0$ we are in Step 1, so let $a' = b' \geq 1$; by Step 2 there is a single pure-A block and a single pure-B block. A mixed column with both blocks on the same side of it has a B-vertex (or, in the mirrored case, an A-vertex) that must reach its block by a path crossing the other part's pure block, which blocks both rows, which is impossible. Hence all mixed columns lie between the blocks, the pure-A block starts at column 0, and the pure-B block ends at column $n - 1$ (or the mirror image, which yields the same set of unordered partitions).

Step 4: the mixed run has constant orientation. Let the mixed run occupy columns $\ell, \dots, r$ and suppose a flip occurs; let d be the leftmost mixed column whose orientation differs from that of column ℓ. Up to reflecting the rows, A holds the tops of columns $\ell, \dots, d-1$ and the bottom of column d. Let $\{(1,d), \dots, (1,e)\}$ be the maximal bottom-row run of A-vertices starting at $(1,d)$, with $e \le r$. Its left neighbor $(1,d-1)$ is in B (column $d-1$ has A on top); its right neighbor $(1,e+1)$ is in B (either a mixed column with A on top, by maximality, or the pure-B block); and every vertical neighbor $(0,j)$, $d \le j \le e$, is in B. So this run is a connected component of A not containing the pure-A block, contradicting connectivity.

Steps 1–4 leave exactly the pattern [pure-A]^a[mixed, orientation σ]$^{n-2a}$[pure-B]^a: this is $S_{a,\sigma}$ when $n - 2a \ge 1$, the vertical split when $n = 2a$, and the horizontal split when $a = 0$. Counting: $1 + 2(\lceil n/2 \rceil - 1) + \mathbf{1}_{\{n \text{ even}\}} = n$ in both parities. □

[DONE] *Proposition 6.5 is proved in full and matches the exhaustive machine counts for all $n \le 12$ (Section 7, item 6). It also corrects an imprecision in an earlier version of this section: the balanced states are staircases, not column cuts; the vertical cut exists only for even n.*

Proposition 6.6 (Stationary law on the strip, $k = 2$)**.** *Let t_a denote the number of spanning trees of the $2 \times a$ grid: $t_1 = 1$, $t_2 = 4$, $t_a = 4t_{a-1} - t_{a-2}$. Under the uniform distribution on balanced 2-forests of the $2 \times n$ grid (equivalently, π_{sp} conditioned on exact balance), the projected law on the n partitions of Proposition 6.5 is*

$$\bar{\pi}(\text{horizontal}) \propto 1, \qquad \bar{\pi}(S_{a,\sigma}) \propto t_a^2, \qquad \bar{\pi}(\text{vertical}) \propto t_{n/2}^2.$$

In particular $\bar{\pi}$ is log-linear in the staircase depth a with rate $2\log(2+\sqrt{3})$ and concentrates on the deepest cuts.

Proof. By Theorem 4.1 the fiber of ξ contains $\prod_i t(G[\xi_i])$ forests. The parts of the horizontal split are paths, each with one spanning tree. The induced graph of $P_{a,\sigma}$ is the $2 \times a$ grid with a path of $n - 2a$ vertices attached by a single edge at a corner; the attached path consists of bridges, present in every spanning tree, so $t(G[P_{a,\sigma}]) = t_a$, and the complementary part is isomorphic to its mirror image, giving t_a^2. The recurrence and growth constant $2 + \sqrt{3}$ (largest root of $x^2 = 4x - 1$) are classical for strip graphs. □

[DONE] *Proposition 6.6 is proved in full. Together the two propositions complete Theorem 6.2 step 1 for $k = 2$ and supply the tree-count calculus (district count $= t_{\mathrm{width}}$, since staircase fringes attach as pendant paths at corners) used for the conditionals in step 2.*

Lemma 6.7 (Split counts and constant split probability on windows)**.** *Let $w \ge 2$, let T be a uniform spanning tree of the $2 \times w$ grid, and let $\xi = \{A, B\}$ range over the balanced connected 2-partitions of the window (Proposition 6.5).*

(a) The spanning trees with $|T \cap \partial\xi| = 1$ (equivalently, those containing an edge whose removal yields exactly ξ, necessarily unique) number exactly $t(G[A]) \cdot t(G[B]) \cdot |\partial\xi|$.

(b) For w even,

$$\Pr[T \text{ admits a balanced split}] \;\geq\; \frac{2\,t_{w/2}^2}{t_w} \;=\; \frac{1}{\sqrt{3}} \cdot \frac{\lambda^w + \lambda^{-w} - 2}{\lambda^w - \lambda^{-w}} \;\geq\; \frac{1}{2}, \qquad \lambda = 2 + \sqrt{3},$$

with equality at $w = 2$ and limit $1/\sqrt{3}$ as $w \to \infty$.

(c) A tree admits at most one balanced split edge per partition, hence at most w balanced split edges in total. Consequently the merge–split proposal (a uniform spanning tree of the window followed by a uniform choice among its balanced split edges) proposes each balanced ξ with probability at least $t(G[A])\,t(G[B])\,|\partial\xi|/(w\,t_w)$, which is within a factor $\mathrm{poly}(w)$ *of the exact conditional $\bar{\pi}(\xi) \propto t(G[A])\,t(G[B])$: the move is a* $\mathrm{poly}(w)$*-accurate independence sampler for the interface conditional.*

Proof. (a) If $T \cap \partial\xi = \{e\}$, then $T \setminus \{e\}$ is an acyclic subgraph of $G[A] \sqcup G[B]$ with $2w - 2$ edges, i.e. the disjoint union of a spanning tree of $G[A]$ and one of $G[B]$; conversely every triple (T_A, T_B, e) with $e \in \partial\xi$ assembles to such a spanning tree, and the two maps are mutually inverse. Removing an edge e' of a spanning tree produces the partition into the two components of $T \setminus \{e'\}$; this equals ξ if and only if $T \cap \partial\xi = \{e'\}$. So the counted trees are exactly those admitting a split at ξ, with a unique split edge.

(b) Apply (a) to the vertical cut: $G[A] \cong G[B] \cong 2 \times (w/2)$ grid and $|\partial\xi| = 2$, giving the lower bound $2t_{w/2}^2/t_w$. With $t_a = (\lambda^a - \lambda^{-a})/(2\sqrt{3})$ and $\lambda\lambda^{-1} = 1$, the middle expression follows by expanding the square. The inequality $\geq 1/2$ is equivalent to $(1 - \frac{\sqrt{3}}{2})\lambda^w + (1 + \frac{\sqrt{3}}{2})\lambda^{-w} \geq 2$, which holds with equality at $w = 2$ and whose left side is increasing in $w \geq 2$ (its derivative has the sign of $(1 - \frac{\sqrt{3}}{2})\lambda^w - (1 + \frac{\sqrt{3}}{2})\lambda^{-w}$, positive already at $w = 2$).

(c) Uniqueness per partition is (a); Proposition 6.5 bounds the number of partitions by w. If a tree has $s \leq w$ balanced split edges, each is chosen with probability $1/s \geq 1/w$; combining with (a) gives the proposal bound. Since $2 \leq |\partial\xi| \leq w + 2$, the proposal law and $\bar{\pi}$ differ by factors in $[\,2/(w(w+2)),\,1\,] \cdot \mathrm{const}$. □

[DONE] *Lemma 6.7 is proved in full and machine-verified: the counting identity of (a) holds exactly for every balanced partition at $w = 4, 6, 8$ (exhaustive enumeration over 56, 780, 10,864 trees); the splittability probabilities are $0.786, 0.764, 0.763$ against the bound $0.571, 0.577, 0.577$; and the measured proposal/target ratios lie in $[0.86, 3.5]$. This retires the estimate previously flagged as the technical heart of Theorem 6.2: what remains of step 2 is to extend the lemma from rectangular windows to the merged windows of* interior *district pairs, whose shape is a ladder with a pendant staircase fringe at each end; the tree-count calculus is unchanged, as pendant fringes contribute factor 1.*

The general classification. For $k \geq 3$ the same column analysis settles the full structure. Write $s = 2n/k$, and set $\hat{t}_0 = 1$ and $\hat{t}_a = t_a$ for $a \geq 1$ (the ladder counts of Proposition 6.6; a district with no full column is a path and has one spanning tree).

Proposition 6.8 (Classification, general k)**.** *Let $k \geq 2$ and $k \mid 2n$. The balanced connected k-partitions of the $2 \times n$ grid are exactly the configurations produced as follows. Choose integers*

$a_1, \dots, a_k \geq 0$, set $\mu_0 = \mu_k = 0$ and $\mu_i = s - 2a_i - \mu_{i-1}$ for $1 \leq i \leq k-1$, and require $\mu_i \geq 0$ for all i together with $2a_k = s - \mu_{k-1}$. The column word, read left to right, is

$$P_1^{a_1} R_1^{\mu_1} P_2^{a_2} R_2^{\mu_2} \cdots R_{k-1}^{\mu_{k-1}} P_k^{a_k},$$

where P_i denotes a full column of district i and each nonempty run $R_j^{\mu_j}$ consists of mixed columns carrying districts j and $j+1$ in its two rows, with a single orientation σ_j throughout the run. Orientations are arbitrary subject to one rule: a district i with $a_i = 0$ lies in a single row, so when both of its runs are nonempty, σ_{i-1} and σ_i must place it in the same row. Distinct admissible codings produce distinct partitions, with one exception: when $a_i = a_{i+1} = 0$ and $\mu_{i-1} = \mu_{i+1} = 0$ (a co-spanning pair*, horizontally splitting the same s columns), the two choices of σ_i give the same partition. Consequently:*

(a) $|\mathcal{B}_k^0(2 \times n)| = \Theta_k(n^{k-1})$;

(b) all interface parities are determined: $\mu_1 \equiv s$ and $\mu_{j+1} \equiv s - \mu_j \pmod 2$;

(c) $t(G[\xi_i]) = \hat{t}_{a_i}$ for every district, so

$$\bar{\pi}(\xi) \;\propto\; \prod_{i=1}^{k} \hat{t}_{a_i} \;=\; \Theta_k(\lambda^{-\sum_j \mu_j}), \qquad \lambda = 2 + \sqrt{3},$$

and the projected measure concentrates exponentially on minimal interface widths.

Informally, a balanced map of the strip is a sequence of solid blocks joined by slanted interfaces. Its probability depends only on the interface widths, and each additional cell of width divides the probability by roughly $\lambda \approx 3.73$.

Proof. Supports are intervals. If district D meets columns $c < c'$ but not an intermediate column d, a path inside D joining the two sides would use a cell of column d, which D does not have.

Overlap structure. A column has two cells, so it meets at most two districts, and two districts sharing a column each occupy one of its rows there. Order the districts by the left endpoints of their supports.

The full columns of a district are consecutive. Suppose district D has full columns $c < c'$ and some intermediate column is not full for D (a *pocket*). Since supports are intervals, D meets every pocket column; the remaining pocket cells belong to districts whose supports lie strictly inside the pocket, as they cannot contain the full columns c, c'. Exactly as in Step 2 of Proposition 6.5, connectivity of D across the pocket forces one entire row of the pocket into D, so the other pocket row is a disjoint union of trapped districts, each of them a path of s cells; the pocket width is then at least s and $|D| \geq 2 + 2 + s > s$, contradicting balance.

Runs and orientations. Between two consecutive full blocks every column is mixed, and by the support ordering the mixed pairs appear as $(i, i+1), (i+1, i+2), \dots$ in left-to-right order; the intermediate districts have no full column, hence contain no vertical edge, hence lie in a single row. Within one maximal run the orientation is constant: a flip creates a maximal one-row segment of one district whose horizontal neighbors on both sides and all vertical neighbors belong to the other district of the run, stranding it from the rest of its district exactly as in Step 4 of Proposition 6.5; for a row-confined district a flip contradicts row-confinement outright.

Balance and the converse. District i has $2a_i$ cells in its full block and one cell in each column of its adjacent runs, giving $2a_i + \mu_{i-1} + \mu_i = s$, which determines the μ_i recursively. Conversely, every admissible coding produces a balanced connected k-partition: each district is a full block with pendant fringes, or a single-row path. The column word determines the partition, and the word determines the coding except that flipping a σ_i can at most permute district names, which yields the same unordered partition exactly in the co-spanning case. For the count, $(a_1, \dots, a_{k-1})$ ranges over the lattice points of a $(k-1)$-dimensional polytope with side lengths $\Theta(s)$, and orientation multiplicities lie in $[1, 2^{k-1}]$. The parity claims follow from the balance equations read from the left end ($\mu_0 = 0$); the tree counts from the pendant-path argument of Proposition 6.6; and the product form from $\sum_i a_i = n - \sum_j \mu_j$ together with $\hat{t}_a = \Theta(\lambda^a)$. □

[DONE] *Proposition 6.8 is machine-verified as a set equality (coded family versus brute-force enumeration) for $k = 2, 3, 4$ at all admissible $n \le 12$, $k = 5$ at $n \in \{5, 10\}$, and $k = 6$ at $n = 9$, together with $t(G[\xi_i]) = \hat{t}_{a_i}$ for every district of every partition (Section 7, item 7). This completes Theorem 6.2 step 1 in full, for exact balance.*

Proposition 6.9 (Interior windows)**.** *Let $\xi \in \mathcal{B}_k^0(2 \times n)$ and let $\{D, D'\}$ be any edge of Q_ξ. Then $G[D \cup D']$ is a $2 \times w$ ladder, $0 \le w \le s$, with a pendant path attached at a prescribed corner on each side (the cells the pair contributes to runs shared with other districts; for non-consecutive Q_ξ-neighbors the union is a ladder with an extended pendant, or a single path when $w = 0$), and $t(G[D \cup D']) = \hat{t}_w$. Writing p, q for the pendant lengths ($p + q + 2w = 2s$), the balanced connected 2-partitions of the window are exactly the staircase splits*

$$A = (\textit{left pendant}) \cup (\textit{first } x \textit{ ladder columns}) \cup (\textit{a fringe of } f \textit{ cells in one row}),$$

up to complementation, where $p + 2x + f = s$, $x, f \ge 0$, and $x + f \le w$. In particular pendants are never severed, there are at most $2(w+1)$ balanced splits, and Lemma 6.7(a) applies verbatim. A split with $f \in \{0, 1\}$ always exists, and for $w \ge 1$

$$\Pr[\textit{a uniform spanning tree of } G[D \cup D'] \textit{ admits a balanced split}] \ge \frac{2\hat{t}_x \hat{t}_y}{t_w} \ge \frac{1}{10}, \qquad x + y \ge w - 1,$$

while for $w = 0$ the window is a path and the probability is 1.

Proof. The shape statement is read off the coding of Proposition 6.8: the pair's full blocks and shared run contribute full columns (the ladder), and the cells the pair holds in runs shared with third districts are one-row segments attached at a ladder corner by a single edge (pendant paths); pendant paths do not change tree counts, so $t(G[D \cup D']) = \hat{t}_w$.

For the splits: a balanced part meeting a pendant contains its attachment corner, since a connected set strictly inside a pendant is a path of fewer than s cells (the case $p = s$, where one part equals the whole pendant, is the split with $x = f = 0$). The two parts have interval column supports, and the analysis of Proposition 6.5 applied to the ladder portion yields exactly the staircase family; counting $x \in \{0, \dots, w\}$ and at most two fringe rows gives at most $2(w+1)$ splits.

Existence: set $f \equiv s - p \pmod 2$ with $f \in \{0, 1\}$, and $x = (s - p - f)/2$. Then $x \geq 0$ because $p \leq s$ (district D's balance equation), and $x + f \leq w$ because $q \leq s$. The parts have tree counts $\hat{t}_x$ and $\hat{t}_y$ with $y = w - x - f$, so $x + y \geq w - 1$, and Lemma 6.7(a) gives the bound $2\hat{t}_x\hat{t}_y/t_w$ (a vertical or near-vertical split has at least two boundary edges). Finally, for $x, y \geq 1$ the closed form $t_a = (\lambda^a - \lambda^{-a})/(2\sqrt{3})$ gives

$$\frac{t_x t_y}{t_{x+y}} = \frac{\lambda^{x+y} - \lambda^{x-y} - \lambda^{y-x} + \lambda^{-x-y}}{2\sqrt{3}\,(\lambda^{x+y} - \lambda^{-x-y})} \;\geq\; \frac{1 - 2\lambda^{-2}}{2\sqrt{3}} \;>\; \frac{1}{5},$$

and the ratio equals 1 when $x = 0$ or $y = 0$; combined with $t_{x+y} \geq t_{w-1} \geq t_w/4$ (from the recurrence), the probability is at least $2 \cdot \frac{1}{5} \cdot \frac{1}{4} = \frac{1}{10}$. □

[DONE] *Proposition 6.9 is machine-verified on* 125 *window configurations ($w \leq 6$, $p, q \leq 3$, all corner placements): the staircase family equals the brute-force set of balanced splits, the split identity is exact, and the minimum observed splittability probability is* 0.40 *against the proven bound* 0.10 *(Section 7, item 8). This completes Theorem 6.2 step 2 in full.*

Corollary 6.10 (Irreducibility on strips, all k)**.** *For every $k \geq 2$ with $k \mid 2n$, the projected window-resampling chain on $\mathcal{B}_k^0(2 \times n)$ is irreducible, with diameter at most $2(k - 1)$.*

Proof. Sweep the interfaces left to right. Districts 1 and 2 are always Q_ξ-adjacent: they share vertical edges when $\mu_1 \geq 1$ (which covers every configuration with $a_1 = 0$, since then $\mu_1 = s$), and abut through full columns when $\mu_1 = 0$. By Proposition 6.9 the resample at interface 1 realizes every admissible $(a_1', \mu_1', \sigma_1')$; choose μ_1' minimal for its parity (0 or 1). Inductively, once interface $j - 1$ has minimal width $\mu_{j-1}' \leq 1$, districts j and $j + 1$ are adjacent: if $\mu_j \geq 1$ they share a run, and if $\mu_j = 0$ then $2a_j = s - \mu_{j-1}' \geq s - 1 \geq 1$, so district j has a full block abutting district $j + 1$'s columns. Setting μ_j' minimal is admissible because $\mu_j' \leq \mu_j$ (equal parities), whence $2a_{j+1}' = s - \mu_j' - \mu_{j+1} \geq s - \mu_j - \mu_{j+1} \geq 0$. After one sweep the interface widths equal the parity vector and the a_i are determined; the only remaining freedom is the orientation of width-one runs, and one further resample per interface realizes any orientation. Hence every state reaches a common canonical configuration in at most $2(k - 1)$ moves. □

[DONE] *Corollary 6.10 settles the strip case of the irreducibility program for* all *k under* exact *balance (compare Conjecture 6.1, which concerns rectangles). Both heat-bath projected chain was also verified irreducible by machine at every tested (k, n).*

Remark 6.11 (Tree-based kernels)**.** *Practical redistricting chains propose window moves through spanning trees, and two cautions emerged from our audit; both are reasons the results above are stated for the window resampling kernel. First, the idealized tree proposal (a uniform spanning tree of the window, then a uniform balanced split edge) admits no closed-form Metropolis–Hastings correction, because the reverse-proposal probability averages the reciprocal number of split edges over all window trees. Second, the implementable corrections condition the proposal tree to pass through the current split, and any such kernel can move between two splits only if some window tree contains exactly one boundary edge of each. Under exact balance the splits of a window cross one another, and exhaustive computation shows that no such tree exists at $s = 4, 6, 8$; the projected tree-based chain is then frozen even though each of its updates is exactly reversible (Section 7, item 11). Chains whose state carries the forest itself, such as the cycle walk and the balanced up-down walk [10, 1], do not face this issue, because the tree is part of the state rather than an auxiliary draw.*

Status of the proof of Theorem 6.2. All four steps are complete. (1) *Structure*: Proposition 6.8. (2) *One interface*: Proposition 6.9, which also makes the heat-bath update explicitly computable, with Corollary 6.10 providing irreducibility. (3) *Many interfaces*: Lemmas 6.12 and 6.13. (4) *Assembly*: the proofs at the end of Section 6.4.

Exact spectral gaps. The random-scan heat-bath projected chain was diagonalized exactly on the coded state space:

k	n	states	$\gamma(P)$
3	6	9	0.400
3	12	33	0.492
3	18	73	0.499
3	24	129	0.500
4	8	25	0.238
4	12	77	0.306
4	16	177	0.326
5	10	64	0.168
5	15	288	0.227

The gap *increases* with n toward $1/(k-1)$, the maximum possible for a random-scan chain on $k-1$ coordinates, and the chain is reversible to machine precision and irreducible at every tested (k, n).

6.4 Proof of the strip gap

In the coordinates of Proposition 6.8 the heat-bath chain admits a simple description. Pick one of the $k-1$ interfaces at random and redraw its width and orientation from the correct conditional distribution, leaving everything else unchanged. The conditional charges a factor of about $\lambda \approx 3.73$ per cell of width, so it is heavily biased toward the narrowest admissible widths. The next lemma makes this quantitative in two ways, and neither statement cares what the rest of the partition looks like.

Lemma 6.12 (Concentration and merging of the conditionals)**.** *Consider the heat-bath update at an interface whose two neighboring interfaces currently have widths u and v, and let $\varepsilon \in \{0, 1\}$ be the parity of the updated width (Proposition 6.8(b)).*

(a) ***Concentration.*** *For every u, v and every $r \geq 0$, the new width m satisfies $\Pr[m \geq \varepsilon + 2r] \leq 31\,\lambda^{-2r}$.*

(b) ***Merging.*** *If $s \geq 8$ and $u, v, u', v' \leq s/4$ (with the parities they must have), the conditionals given (u, v) and given (u', v') differ in total variation by at most $40\,\lambda^{-s/4}$.*

Proof. Up to a factor not depending on m, the weight of width m is $\lambda^{-m} g(m)$ with $g(m) = h(a)\,h(a')\,\nu(m)$, where $a = (s-u-m)/2$ and $a' = (s-m-v)/2$ are the two block widths, $h(a) := \hat{t}_a/(A\lambda^a)$ with $A = 1/(2\sqrt{3})$, and $\nu(m) \in [1, 2]$ counts the admissible orientations. From the closed form, $h(a) = 1 - \lambda^{-2a} \in [1 - \lambda^{-2}, 1)$ for $a \geq 1$, and $h(0) = 2\sqrt{3}$. Hence $g(m) \in [c_-, c_+]$ with $c_+/c_- \leq 2\,(2\sqrt{3})^2/(1-\lambda^{-2})^2 \leq 28$.

(a) The smallest admissible width is ε whenever the support is nonempty, so

$$\Pr[m \geq \varepsilon + 2r] \;\leq\; \frac{c_+ \sum_{i \geq r} \lambda^{-(\varepsilon+2i)}}{c_- \lambda^{-\varepsilon}} \;=\; \frac{c_+}{c_-} \cdot \frac{\lambda^{-2r}}{1-\lambda^{-2}} \;\leq\; 31\,\lambda^{-2r}.$$

Truncating the support from above only helps, and the degenerate point-mass cases are trivial.

(b) On widths $m \leq s/2$ both conditionals are supported (the largest admissible width is at least $3s/4$), both blocks satisfy $a, a' \geq s/8 \geq 1$, so no orientation constraint is active and ν is the same, while the factors $1-\lambda^{-2a}$ of the two conditionals differ by a multiplicative $1 \pm 3\lambda^{-s/4}$. Widths above $s/2$ carry mass at most $31\lambda^{-s/2+2}$ under either conditional, by (a). A ratio comparison on the common part plus the two tails gives total variation at most $40\,\lambda^{-s/4}$. □

Lemma 6.13 (Coupling of the interface chain)**.** *There are absolute constants $C_0, c > 0$ such that for every $k \geq 2$ and every $s \geq C_0 \log(k+1)$, the random-scan heat-bath interface chain satisfies*

$$t_{\mathrm{mix}}(P^{\mathrm{HB}}, 1/4) \;\leq\; 2(k-1)(\lceil \log(k+1) \rceil + 4), \qquad \gamma(P^{\mathrm{HB}}) \;\geq\; \frac{c}{k \log(k+1)}.$$

Proof. Run two copies X and Y from arbitrary states. At every step both copies update the *same* uniformly chosen interface, and the two new (width, orientation) pairs are drawn from the maximal coupling of the two conditionals. Let $M = \lfloor s/4 \rfloor$, let each of phase 1 and phase 2 last $T = (k-1)(\lceil \log(k+1) \rceil + 4)$ steps, and consider three events.

First, in each phase every interface is selected at least once; missing one is a coupon-collector event of probability at most $(k-1)e^{-T/(k-1)} \leq e^{-4}$ per phase.

Second, every update in either phase lands at width at most M in both copies. A single update fails with probability at most $31\lambda^{-M+1}$ by Lemma 6.12(a), whatever the neighbors are, so a union bound over at most $4T$ update-copies gives $124\,T\lambda^{-M+1}$. An interface at which the update holds (its window is disconnected, which requires a neighboring interface of width s) already has width $0 \leq M$. Hence after phase 1 every interface width in both copies is at most M, and this persists through phase 2.

Third, every phase-2 update draws the same pair in both copies. Given all widths at most $M \leq s/4$, no update holds (a hold needs a neighbor of width s), and Lemma 6.12(b) bounds the failure of each maximal coupling by $40\lambda^{-s/4}$, so a union bound gives $40\,T\lambda^{-s/4}$.

On the intersection of the three events, every phase-2 update leaves the updated interface equal in the two copies; in particular, after the last update of each interface the copies agree there, and that interface is not modified again. Since every interface is updated during phase 2, $X = Y$ after $2T$ steps. The total failure probability is at most $2e^{-4} + 124\,T\lambda^{-M+1} + 40\,T\lambda^{-s/4} \leq 1/4$ once $s \geq C_0 \log(k+1)$ for an absolute C_0. The coupling inequality gives the mixing bound, and $t_{\mathrm{mix}}(1/4) \geq (t_{\mathrm{rel}} - 1)\log 2$ converts it into the gap bound. □

Proof of Theorem 6.2. The chain $\bar{P}$ is exactly the random-scan heat-bath interface chain, so Lemma 6.13 bounds its mixing time, and $t_{\mathrm{mix}}(1/4) \geq (t_{\mathrm{rel}} - 1)\log 2$ converts the bound into the gap bound. For smaller s the chain is a finite, reversible, irreducible chain (Corollary 6.10), so its gap is positive. □

Proof of Corollary 6.4. By Theorem 5.2, $t_{\text{mix}}(P, \epsilon) = t_{\text{mix}}(\bar{P}, \epsilon)$, and $\bar{P}$ is exactly the random-scan heat-bath interface chain, so Lemma 6.13 gives the mixing bound directly; the coupling bounds total variation itself, so no spectral argument and no laziness are needed. The finitely many n with $s < C_0 \log(k+1)$ contribute a constant by Corollary 6.10, and $k = 2$ is Remark 6.3. For the cost per step, P_F runs Wilson's algorithm in every district and P_M resamples one window, with the split distribution computed from Proposition 6.9 and the ladder recurrence. □

[DONE] *Lemmas 6.12 and 6.13, Theorem 6.2, and Corollary 6.4 are proved in full. Both estimates of Lemma 6.12 are machine-verified with wide margins: the tail constant has a 33% cushion over all neighbor widths at $s \leq 40$, and the measured merging total variation is three or more orders of magnitude below the stated bound (Section 7, item 10).*

[NOTE] *Scope: the strip program is complete for exact balance; the ε-balanced variant of the classification is routine but not written. Torpidity examples of [7] live on grid-*like *planar families and do not touch strips; any claim beyond constant-height strips must engage the geometry of [5] seriously. Square grids are stated as the open problem, not promised.*

7 Machine verification

All computations are exhaustive (no sampling) and were performed in exact integer or rational arithmetic. Scripts are available from the author.

1. **Torsor axioms (Theorem 3.1).** For the 2×3 grid with S a pair of opposite corners ($|\mathcal{F}_S| = 21$) and with a root triple (13), the 3×3 grid with opposite corners (288), and K_4 with an adjacent pair (8; here G_S has parallel edges): every generator acts as a permutation of $\mathcal{T}(G_S)$; the chip-firing relations hold, so the action factors through $\text{Pic}_k(G)$; generators commute; and the orbit of a single tree has full size. Freeness follows from transitivity and $|\text{Pic}_k(G)| = |\mathcal{F}_S(G)|$.

2. **Matrix-forest consistency.** For *every* root set of every test graph (triangle, C_4, K_4, 2×3, 3×3; 162 root sets in total): $\det \Delta^{(S)} = |\mathcal{F}_S(G)| = t(G_S)$, and the Smith normal forms of $\Delta^{(S)}$ and of the reduced Laplacian of G_S agree.

3. **Fiber decomposition (Theorem 4.1).** Verified by enumeration on five partitions of the 2×3 and 3×3 grids ($\tau = 3, 4, 8, 45, 32$) and on a 6×6 quadrant instance ($\tau = 192^4 \cdot 108$).

4. **No algebraic shortcut for balance.** On the 2×3 grid with $S =$ opposite corners: of the 9 partition fibers in $\text{Pic}_k(G) \cong$ (21 elements), none of the fibers of size > 1 is a coset of a subgroup, and the union of the three balanced fibers is not a coset. Moreover no group homomorphism $\text{Pic}^0(G) \to \text{Pic}_k(G)$ can exist in general ($|\text{Pic}_k()| \nmid t(G)$ for 162 of the tested root sets, including the triangle with $k = 2$: $2 \nmid 3$).

5. **Irreducibility evidence.** On the 3×3 grid, $k = 2$, no balance constraint: the boundary-restricted single-swap chain on (tree, marked-edge) states is irreducible: all 1,536 states form one class covering all 53 connected 2-partitions. With exact balance ($k = 3$, districts of size 3): the state graph has two isolated states, but the main class covers all 10 balanced

partitions, consistent with Conjecture 6.1 at the partition level and a warning at the lifted level.

6. **Balanced partition counts on strips.** Exhaustive counts of $|\mathcal{B}_k^0(2 \times n)|$: for $k = 2$, exactly n for all $n \leq 12$, matching Proposition 6.5; for $k = 3$: $3, 9, 19, 33$ at $n = 3, 6, 9, 12$; for $k = 4$: $5, 11, 25, 45, 77$ at $n = 4, 6, 8, 10, 12$, consistent with $\Theta(n^{k-1})$ growth (constant second differences at $k = 3$, cubic growth at $k = 4$).

7. **General classification (Proposition 6.8).** The coded family equals the brute-force set of balanced connected k-partitions, as sets of partitions, for $k = 2, 3, 4$ at all admissible $n \leq 12$, $k = 5$ at $n \in \{5, 10\}$ (8 and 64 partitions), and $k = 6$ at $n = 9$ (41 partitions); and $t(G[\xi_i]) = \hat{t}_{a_i}$ holds for every district of every coded partition (matrix-tree check). The balanced-forest multiplicities on $(k, n) = (3, 6)$ equal $\prod_i \hat{t}_{a_i}$ for all 9 partitions, confirming the projected law.

8. **Interior windows (Proposition 6.9).** On 125 pendant-ladder configurations ($w \leq 6$, $p, q \leq 3$, all corner placements): the staircase family equals the brute-force set of balanced splits, the split-count identity is exact for every split, $t(H) = \hat{t}_w$, and the minimum splittability probability is 0.400 (proven bound 1/10). The pendant-free rectangular cases ($w = 4, 6, 8$; 56, 780, 10,864 trees) give splittability 0.786, 0.764, 0.763 against the limit $1/\sqrt{3} \approx 0.577$.

9. **Projected spectral gaps.** Exact diagonalization of the heat-bath chain at the (k, n) pairs of the table in Section 6.3: reversible to machine precision, irreducible everywhere, and $\gamma(\bar{P}) \to 1/(k-1)$ as n grows.

10. **Coupling estimates (Lemma 6.12).** Exact conditionals at all admissible neighbor widths for $s \leq 40$: the tail bound holds with maximum ratio 0.67 of the stated constant, and the merging total variation at $s = 16, 24, 32, 40$ is smaller than $\lambda^{-s/4}$ by three or more orders of magnitude.

11. **Tree-based kernel freeze (Remark 6.11).** The through-the-current-split merge–split kernel satisfies detailed balance to machine precision, yet its off-diagonal support is empty at every tested size: at (k, n) with $s = 4, 6, 8$ the projected chain has spectral gap exactly 0 and is reducible.

[DONE] *All five computations were run to completion in July 2026.* **[NOTE]** *Before submission: re-run the suite from a clean repository, commit the scripts (currently in a scratch directory), and add a 4×4, $k = 2$ instance of item 5, which is within reach ($t(4 \times 4) = 100{,}352$ trees; restrict to balanced marked states).*

8 Discussion

The division of labor proposed here is strict: algebra handles everything that is exactly solvable (uniformity on fibers, torsor navigation, within-fiber waste, which Lemma 5.1 eliminates rather than mitigates), and the remaining difficulty is a spectral gap estimate for an explicit

chain on an explicit nearly-one-dimensional state space (Theorem 6.2, now proved on strips). We emphasize what the collapse lemma does *not* do: it does not make the projected chain fast: torpidity examples [7] project too. It makes the projected chain the *whole* problem, which no existing analysis of a balanced chain achieves. With the strip program complete, genuine two-dimensional grids are the central open problem.

Two further directions are natural. First, the multi-sink torsor suggests global P_M moves: resample the entire forest conditioned on a root set via Wilson on G_S, with roots re-drawn from a suitable distribution; the acceptance ratio is computable by Theorem 4.1, and such moves may shortcut the interface-by-interface diffusion on strips. Second, the merge–split ledger of Xu's partition divisors (the minimal effective divisor D^ξ with $\deg D^\xi = |\partial\xi|$, and the merge formula of [20, Cor. 3]; see also the resolution theorems of [15, 17]) provides an exact algebraic bookkeeping of merge–split paths that may serve canonical-path constructions on general planar graphs.

[**MISSING**] *Decide, after Theorem 6.2 is resolved, whether the global-move variant belongs in this paper or the next one.*

A Status

Item	Status
Thm 3.1 (multi-sink torsor, (i)–(iii))	Proved in full; machine-verified.
Rem. 3.2 (multi-sink Dhar)	Immediate from Thm 3.1; verified.
Cor. 3.4(a) (Wilson exact sampling)	Proved (citation to Wilson '96).
Cor. 3.4(b) (navigation)	Proved as stated; torsor discrete-log deliberately avoided.
Thm 4.1 (fiber decomposition)	Proved in full; verified numerically.
Prop. 4.2 (exact fiber resampler)	Proved in full.
Lem. 5.1 (fiber collapse, (a)–(d))	Proved in full.
Thm 5.2 (conditional mixing theorem)	Proved in full (conditional on projected gap).
P_M specification, MH correction details (§6.1)	Write out the exact proposal/acceptance formulas; routine but must be in the paper.
Conj. 6.1 step 1 ($k = 3$ rectangles)	Open; adapt [1, App. B]; first priority.
Conj. 6.1 steps 2–3 (general k)	Open; competition risk (BUD authors).
BUD $\to$ merge–split transfer ($k = 2$ base case)	Believed routine via [1, Lem. A.3–A.5]; write it.
Thm 6.2 step 1, $k = 2$: classification (Prop. 6.5, exactly n staircases) and stationary law (Prop. 6.6)	Proved in full; counts machine-verified to $n = 12$.
$k = 2$ triviality of merge–split (Rem. 6.3); Theorem 6.2 restated for $k \geq 3$	Done.
Thm 6.2 step 1, $k \geq 3$: general classification, parities, product-form stationary law (Prop. 6.8)	Proved in full; set equality machine-verified for $k \leq 6$.
Strip irreducibility for all k, exact balance, diameter $\leq 2(k-1)$ (Cor. 6.10)	Proved in full.
P_M implementability audit: tree-based MH kernels frozen under exact balance; theorem chain switched to window heat-bath (Rem. 6.11)	Done; freeze machine-verified at $s = 4, 6, 8$.
Thm 6.2 step 2 on rectangular windows: split identity, constant split probability, poly-accurate proposal (Lem. 6.7)	Proved in full; machine-verified $w = 4, 6, 8$.
Thm 6.2 step 2 on interior windows: shapes, split classification, splittability $\geq 1/10$ (Prop. 6.9)	Proved in full; verified on 125 configurations.
Thm 6.2 step 3: concentration + coupling (Lems 6.12, 6.13)	Proved in full; estimates machine-verified with wide margins.
Thm 6.2 and Cor. 6.4: unconditional strip mixing, $O(k \log k)$ steps, near-linear total time	Proved in full: the headline theorem.
Exact gap diagonalization ($k = 3, 4, 5$; $n \leq 24$)	Done; table in §6.3.
4×4 balanced irreducibility computation	Not yet run; feasible.
Scripts committed to repository	Currently in session scratch directory.
Related-work completeness pass (incl. any 2026 preprints)	Re-sweep arXiv before submission.

Reading order for finishing the paper. (1) Theorem 6.2 step 2 is the pivot: if the one-interface split-probability bound works out on strips, everything else is assembly and the paper's headline becomes Corollary 6.4 unconditionally for $k = 2$; submit to SODA/STOC. (2) If step 2 stalls, the fallback paper is Sections 3–5 plus Conjecture 6.1 step 1 proved (irreducibility, $k = 3$ rectangles, exact balance): a coherent algorithms paper for ESA/SODA. (3) The infrastructure alone (Sections 3–4) is the minimal publishable unit, as a combinatorics note.

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